

YAGNYA MURTI POKHREL

Lubbock, Texas-79415 || ympokhrel@gmail.com || +1-806-560-2409 || [[Linkedin](#)]

SUMMARY

- 7+ years of experience in digital/analog circuit design, PCB development, VLSI, and embedded systems.
- Proficient in hardware testing and troubleshooting using lab instruments (oscilloscopes, spectrum analyzers, NI myRIO, LabVIEW).
- Proficient in schematic capture and PCB design using Altium Designer, with hands-on experience in FPGA-based hardware development and system integration for embedded and mixed-signal applications.
- Research experience in material behavior, electron field emission, and electron beam and laser surface treatment techniques for vacuum electronics and HPM devices.

EDUCATION

PhD, Electrical Engineering , Texas Tech University, USA	Aug 2022 – May 2025	(GPA: 4.0)
MS, Electrical Engineering , Texas Tech University, USA	Aug 2022 - May 2024	(GPA: 4.0)
MS, Information & Communication , Tribhuvan Univ., Nepal	Nov 2010 - Oct 2012	(GPA: 4.0)
BE, Electronics and Communication , Tribhuvan Univ., Nepal	Aug 2004 - Dec 2008	(GPA: 3.9)

SKILLS

- **Programming:** Python, MATLAB, C, C++, Fortran 90,
- **Software Tools:** Altium Designer, NI LabVIEW, Cadence, SPICE, LogixPro-PLC, LTspice, MathCAD, LAMMPS, SQL, SRIM, COMSOL, HPCC, Power world, PSS/E, SAM, Simulink, JAM
- **Hardware Description Languages:** Verilog, VHDL
- **Testing:** Oscilloscopes, Logic Analyzers, Network Analyzers, NI myRIO
- **Lab Equipment:** DMMs, RF Signal Generators, Spectrum Analyzers

CERTIFICATIONS & TRAININGS

- **IBM Data Analyst Professional Certificates** (7 Courses), Coursera: [CERTIFICATIONS](#)
- **Semiconductor Fabrication 101** (January 2025), Purdue University: [CERTIFICATIONS](#)
- **PCB Design Training:** Altium Education, November 2024
- **Critical Infrastructure Security** Training, Texas Tech University, 16 hours, July 2023.

RELEVANT WORK EXPERIENCES

Research Assistant, Texas Tech University, Lubbock TX August 2022 - Present

- Performed high-level computational modeling and multiphysics simulations using tools such as COMSOL, LAMMPS, MATLAB, and KMC, focusing on electron emission, beam dynamics, laser surface processing, and material stability in HPM sources, vacuum electronics, and nanoelectronic devices. Integrated and validated findings with experimental data to enhance accuracy and reliability.

Technical Leader, Techno Village Pvt. Ltd., Nepal February 2017 – June 2022

- Led manufacturing, testing, and installation of electronic products (RFID & PIN-Based Bank Locker Systems, 12V DC Power Supplies, Digital Display Systems, Automated Control Panels, IGBT-Based ELCs, AVR), enhancing switching efficiency and reliability.

RELEVANT GRADUATE PROJECT

- **Testing of XOR (SN74HC86) Gate using NI myRIO**
- Conducted continuity, functionality and parametric tests using NI myRIO and LabVIEW, ensuring device compliance with datasheet specifications and performing statistical analysis (Cp, Cpk).

HONORS & AWARDS

- **Kul Ratna Tuladhar, Gold Medal** award for topping Electronics Engineering diploma in Nepal, 2003.
- Member, **Phi Kappa Phi Honor Society & Tau Beta Pi Honor Society**, USA.
- **Golden Jubilee Scholarship** 2004-05, Government of India, for B.E. in Nepal.

PUBLICATIONS

- ⇒ Authored 6 research papers on advanced materials, electron emission, heating mechanisms, laser surface treatments, and methods for minimizing outgassing in HPM systems and vacuum electronics.

PROFESSIONAL LICENSES

- ⇒ FE Electrical and Computer Engineering, Texas Board, 08/15/2023. NCEES ID: 23-735-75 [LINK](#)